

Shervin Faramarzi Babadi

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EDUCATION

M.Sc., TH Köln University of Applied Sciences <i>Master of Science in Automation & Information Technology (IT)</i>	Cologne, Germany 2026 – Present
M.Sc., Azad University Najafabad Branch <i>Master of Science in Electrical Engineering (Control Systems)</i> Thesis Title: <i>Adaptive Fuzzy Iterative Learning Control for Quadrotor Path Tracking</i> cGPA: 17.43 (out of 20), German equivalent: 1.8 Advisor: Dr. Abbas Chatraei	Isfahan, Iran 2015 – 2019
B.Sc., Azad University Najafabad Branch <i>Bachelor of Science in Electrical Engineering (Control Systems)</i> Thesis Title: <i>Performance of Static Var Compensation Systems (SVC) in Steel Companies</i> cGPA: 16.05 (out of 20), German equivalent: 2.2 Advisor: Dr. Homayoun Mahdavi-Nasab	Isfahan, Iran 2011 – 2015

FIELDS OF INTEREST

IT & Data: Machine Learning, Data Science, Data Analytics, Automation
Digitalization: Monitoring Systems, Data Acquisition, Industrial IoT
Control Systems & Robotics: Adaptive Control, Fuzzy Control, Optimal Control, Motion Control
Signal Processing: Digital Signal Processing, Audio Signal Analysis
Energy Systems & Analytics: Load Monitoring, Load Forecasting, Energy Data Analysis

PUBLICATIONS

P1 Shervin Faramarzi, Abbas Chatraei, “*Adaptive Fuzzy Iterative Learning Control for Quadrotor Path Tracking*”, 5th National Conference on Electrical Engineering of Iran, Najafabad, Iran, 2018.

ACADEMIC EXPERIENCE

Azad University Najafabad Branch, Isfahan, Iran <i>Teaching Assistant — Linear Control Systems</i> Assisted in teaching classical control methods (root locus, Bode plots, PID) using MATLAB/Simulink. Supported students in modeling and analysis of dynamic systems.	Fall 2013
Azad University Najafabad Branch, Isfahan, Iran <i>Teaching Assistant — System Identification</i> Assisted in lectures and labs on parameter estimation, model validation, and identification algorithms using MATLAB/Simulink; evaluated assignments and provided feedback.	Fall 2017

EMPLOYMENT

Makan System Daneshmand Company, Isfahan, Iran <i>Programmer & Data Analyst</i> Developed Python workflows to clean and standardize unstructured technical notes using AI-assisted tools where necessary. Validated results through post-processing to improve documentation usability.	Aug 2023 – Jan 2026
Atiehsazanfarhood Company, Isfahan, Iran <i>Research & Development Engineer</i> Assisted with documentation and reviewed technical information within the R&D department while developing foundational knowledge in automation and control systems.	Feb 2020 – Aug 2023
Azad University Najafabad Branch, Robotics Institution, Isfahan, Iran <i>Part-time Researcher</i> Researched robotic control systems; modeled and simulated a SCARA robot arm, implemented control algorithms, and optimized motion performance.	2018 – 2019
Mobarakeh Steel Company, Mobarakeh, Isfahan, Iran <i>Intern</i> Completed a 360-hour internship in power distribution. Worked with PLCs and automation systems, assisted in troubleshooting, and gained hands-on experience in industrial power operations.	Feb 2015 – Apr 2015

PROJECTS

[Non-Intrusive Load Monitoring (NILM) — Siemens SENTRON PAC4200 & Machine Learning]

Developed a NILM system using real-world Siemens PAC4200 measurements to identify individual appliances from aggregate electrical data. Engineered power-based features and trained a Random Forest classifier, with visualization for monitoring and validation.

Technologies: Python, Random Forest, Siemens SENTRON PAC4200, Feature Engineering, Data Visualization

[Electricity Load Forecasting — ENTSO-E & LightGBM]

Built a 24-hour German load-forecasting pipeline using ENTSO-E, weather, and calendar data. Engineered time-series features, trained a LightGBM model, and benchmarked performance with backtesting.

Technologies: Python, LightGBM, Time-Series Forecasting, ENTSO-E, Feature Engineering

[Morse Code Audio Decoder — Python, DSP & Adaptive WPM Detection]

Developed a real-time Morse code audio decoder using DSP and adaptive timing analysis. Implemented FFT-based frequency detection, bandpass filtering, Hilbert-envelope extraction, and adaptive WPM estimation for decoding audio from files and microphone input.

Technologies: Python, DSP, FFT, SciPy, Adaptive Signal Processing, Real-Time Audio

AWARDS

Ranked 1st among M.Sc. students in Control Systems, Electrical Engineering Department, Azad University Najafabad.

Ranked 2nd among B.Sc. students in Control Systems, Electrical Engineering Department, Azad University Najafabad.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MATLAB/Simulink, C/C++, Git, Jupyter

Machine Learning: Scikit-learn, LightGBM, Random Forest, Feature Engineering

Data & Visualization: Pandas, NumPy, Matplotlib, Microsoft Excel

Signal Processing: SciPy, FFT, Digital Signal Processing (DSP)

Automation & Industrial: Siemens TIA Portal, PLC/SCADA, Industrial IoT, Node-RED

Control Systems: Adaptive Control, Fuzzy Control, LQR, System Modeling & Simulation

Enterprise: SAP

CERTIFICATES & PROFESSIONAL TRAINING

Capstone: Retrieving, Processing, and Visualizing Data with Python

Jan 2026

University of Michigan — Coursera

Using Python to Access Web Data

Sep 2025

University of Michigan — Coursera

Programming for Everybody (Getting Started with Python)

Jun 2025

University of Michigan — Coursera

Python Data Structures

Aug 2025

University of Michigan — Coursera

LANGUAGE PROFICIENCY

English: Fluent C1 — IELTS Overall Score: 7.0 (Speaking: 7.0, Listening: 8.5, Reading: 7.0, Writing: 6.0)

Deutsch: A2 - currently improving

Farsi (Persian): Native

EXTRACURRICULAR ACTIVITIES

- Swimming, Tennis, Aerobics, Yoga

REFERENCES AVAILABLE TO CONTACT

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